

Creating Quality Training Data

The 4 quality pillars: diversity, correctness, format consistency, length balance

10 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

AI-FT-IN-TH-000012

ID

The 4 Quality Pillars of Fine-Tuning Data

Pillar 1: Diversity

Cover a wide range of tasks, domains, difficulty levels, and response styles. Narrow training data → narrow capability. Measure with entropy over task categories.

Pillar 2: Correctness

Ground-truth outputs must be accurate, especially for factual, code, and math tasks. One wrong example can propagate error patterns throughout fine-tuning.

Pillar 3: Format Consistency

Use a single prompt template throughout. Mixing formats confuses the model — it learns competing format patterns rather than the task itself.

Pillar 4: Length Balance

Mix short (1–2 sentence) and long (paragraph) responses. Imbalanced length distribution → biased generation length. Compute length histogram.

Strategies for Data Selection — Quantity vs Quality

Quality-First (LIMA approach)

- 1,000 carefully curated examples > 50,000 noisy examples
- Manual review of every example for accuracy
- Diversity sampling: ensure coverage of task types
- Perplexity-based filtering: remove trivially easy examples
- LLM-as-judge: use GPT-4 to score output quality

Quantity-First (Web-scale)

- Filter billions of examples with automated heuristics
- MinHash deduplication to remove near-duplicates
- Perplexity thresholding with reference model
- Toxicity classifier (Perspective API) for safety
- Best for pre-training; fine-tuning still benefits from quality

Dataset Construction Checklist — Before Training

Step 1: Format all examples to a single template (e.g. ChatML):

```
assert all(example["format"] == "chatml" for example in dataset)
```

Step 2: Check deduplication (MinHash Jaccard threshold 0.9):

```
python dedup.py --threshold 0.9 --input data.jsonl --output data_dedup.jsonl
```

Step 3: Length distribution — compute histogram:

```
lengths = [len(ex['output']).split()) for ex in dataset]
```

Target: p50 < 200 words, p99 < 1000 words, no zero-length outputs

Step 4: Correctness spot-check — random sample 50 examples, manual review

Step 5: Task diversity — count examples per task category, ensure < 30% from any single category

KEY REFERENCES

- Zhou, C. et al. (2023). LIMA: Less Is More for Alignment. NeurIPS 2023. arxiv.org/abs/2305.11206
- Dubois, Y. et al. (2023). AlpacaFarm: A Simulation Framework for Methods that Learn from Human Feedback. NeurIPS 2023.

 **Next: Concept 16: Multi-Task Instruction Tuning (AI-FT-IN-TH-000013)**